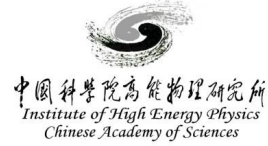


CEPC NOTE

CEPC-CEPCSample-2015-009

February 25, 2019



Generated Sample Status for CEPC Simulation Studies

Mo Xin^a and Li Gang^a

^aInstitute of High Energy Physics, Chinese Academy of Science, Beijing, China

Abstract

This note focus on the event generation for CEPC studies. The signal and background samples are generated by Monte-Carlo generator Whizard and grouped according to their final states, and the cross sections are given out. All the results are compared with ILC experiment.

E-mail address: li.gang@ihep.ac.cn

© Copyright 2019 IHEP for the benefit of the CEPC Collaboration.

Reproduction of this article or parts of it is allowed as specified in the CC-BY-3.0 license.

1 Introduction

The Circular Electron-Positron Collider(CEPC) is a proposed circular electron positron collider experiment to be committed. The main objective is to reconfirm the observation of the Higgs-like particle, which was observed by LHC experiment in July 2012, and study the properties of Higgs boson. The Monte-Carlo event samples play an important role to check the performance of CEPC detector and study the expected branch ratio predicted by SM or BSM of the Higgs boson.

All the samples mentioned in this note are generated by the MC generator Whizard, version 1.95. The samples are grouped into signal and background, and then classified according to their final states, which will be discussed in the following sections. The samples are also compared with ILC, the comparisons are listed in the tables.

Here are some considerations for our samples:

1. The luminosity of the ILC samples is not unified to a definite number, but varies from 250 fb^{-1} to above 1000 fb^{-1} , and for some background samples the luminosity is not high enough. We require our samples at the unified luminosity of 5000 fb^{-1} .

2. There will be some special requirements for some particular process in future analysis. For example, If the analysers want to separate the WW-fusion events from the “ZH” signal(although they are interfered with each other in reality), the fusion sample will be provided.

3. The Beyond Standard Model(BSM) is the main stream for the new physics. Therefore the BSM samples will be provided according to the requirements.

The general information about the samples is listed in Tab.1:

Table 1: The general information about samples

	Luminosity [fb^{-1}]	ISR	Beamstrahlung	Polarization
Signal	5000	On	Off	Both
Background	5000	On	Off	Non-polarized
Path	/cefs/data/stdhep/			

Additional, the curve of the total cross section respective to $\sqrt{s}$ under SM for various processes discussed in this note is given out in Fig.??, and the value at 250GeV for these curves is listed in Tab.2. The “ZH” process is our signal part, the “W fusion” and “Z fusion” correspond to the t -channel Feynman diagrams. The others belong to the backgrounds.

Table 2: The value of the cross section at 250GeV

Process	$q\bar{q}$	$\mu^+\mu^-$	Single Z	Single W	Bhabha
$\sigma[\text{fb}]$	50216.20	4404.69	4733.74	5144.28	25060.22
Process	WW	ZZ	ZH	Z fusion	W fusion
$\sigma[\text{fb}]$	15483.95	1033.38	212.13	0.63	6.72

2 Higgs signal samples

The Higgs signal samples play an important role in the research of Higgs physics. We mainly focus on the process of $e^+e^- \rightarrow ZH$, which will be called as “ZH” process afterwards, then the Higgs and Z

bosons will decay further. In the signal samples, the Higgs decays inclusively and Z decays into visible productions.

However, the final states of the signal samples are not specified as Higgs and Z bosons, but their decaying productions instead. Therefore the samples not only contain the “ZH” process actually, but also contain other processes, typically the W-W fusion process when the final states are neutrino, anti-neutrino and Higgs, like the Fig.1 says. The manipulation in such a way could generate the real-world signal samples and guarantee the corresponding cross sections gauge invariant.

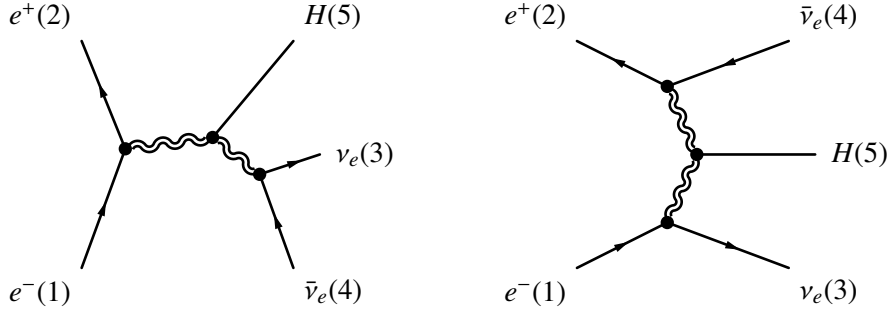


Figure 1: The diagrams for process $e^+, e^- \rightarrow \nu_e, \bar{\nu}_e, h$

The Higgs signal samples are grouped according to the Z decaying productions (although they do not necessarily come from Z) and named according to them. For example, the directory named as “E250.Pn1n1h.whizard195” denotes that the samples are generated at 250GeV c.m. energy with Whizard 1.95, the letter “P” in “Pe1e1h” stands for the word “process”, “n1n1” is the production of Z decay and also the name of the process. Actually it means neutrino and the anti-neutrino, they could come from the W-W fusion process and the samples also contain such events indeed. At last, “h” stands for Higgs obviously.

The samples will be grouped according to the polarizations of the incoming particles even further. There are four distinct combinations for each sample, the incoming electron and positron being assigned with left or right respectively. So you can find four sub directories for every process, for instance, you can find from “E250.Pn1n1h.X.el.pl.whizard195”, to “E250.Pn1n1h.X.er.pr.whizard195”, undergoing all the four combination of the polarization, in the directory “E250.Pn1n1h.whizard195”. The letter “X” denotes that the Higgs decay inclusively, and the sub string between “X” and “whizard195” describes the combination of the polarized incoming particles. Besides these polarised samples, the unpolarised one named as “e0.p0.” could be found at the same location as well.

The Higgs boson will decay according to the SM model inclusively and the production will be hadronized by Pythia if partons exist. And there are some points should be noted:

1. For the processes with small cross section, typically under 20 fb, the total number of event is not enough to make the statistics reliable. Therefore, a lower limit, saying 100000, is necessary for the actual usage. When turned back to the analysis, a scaling factor is needed to get the real statistics. Taking the process “e2e2h_X” as an example, the cross section is just 7.10 fb, the number of events is 35849, which is under the lower limit 100000, the sample will be set to be generated as 100000.

2. However, a small amount of events will not generated eventually due to some technical problem of the generators. To compensate the lost events, the actual luminosity will be set as 5050 fb^{-1} , and the number of events for various processes is also expected under luminosity 5050 fb^{-1} .

3. Finally, the expected events number and the actual one for all the signal and background processes are listed in Tab.17-Tab.19. So the scaling factor for the corresponding processes is calculated

$$scale = \frac{\text{No. of evt expeted}}{\text{No. of evt generated}},$$

which is almost 1, except whose cross section is under 20 fb.

2.1 Leptonic

The leptonic part of the signal samples include the processes of Z decaying into leptons, meanwhile Higgs decaying inclusively. Samples are generated for each type of lepton separately and also there is a directory contains $Z \rightarrow lepton$ as a whole. The information about these samples is divided into electronic and neutrino parts, and given out in Tab.3 and Tab.4 respectively.

Table 3: The cross section for the electronic part of Higgs signal

Process	Cross section [fb]	Error [fb]	ILC result [fb]
e1e1h_X	7.60	0.006	7.19
e2e2h_X	7.10	0.004	7.03
e3e3h_X	7.08	0.004	7.02

Table 4: The cross section for the neutrino part of Higgs signal

Process	Cross section [fb]	Error [fb]	ILC result [fb]
nnh_X	48.96	0.033	48.43
n1n1h_X	20.91	0.017	–
n2n2h_X	14.03	0.008	–
n3n3h_X	14.01	0.008	–

2.2 Hadronic

The hadronic part of the signal samples include the processes of Z decaying into quarks, meanwhile Higgs decaying inclusively. Samples are generated for each flavor (except the top quark) of quarks separately and also there is a directory contains $Z \rightarrow q\bar{q}$ as a whole. The samples will be transferred from the parton level to hadron level by Pythia finally. The information about these samples is given out in Tab.5.

Table 5: The cross section for the hadronic part of Higgs signal

Process	Cross section [fb]	Error [fb]	ILC result [fb]
qqh_X	143.39	0.092	141.99
uuh_X	24.52	0.016	–
ddh_X	31.45	0.020	–
cch_X	24.51	0.016	–
ssh_X	31.46	0.020	–
bbh_X	31.18	0.020	–

3 The two fermions background

There are two different topological structure of two fermions Feynman diagrams. One is in s-channel, like in Fig.2, the other is in t-channel, shown as Fig.2. The t-channel can be classified still further according to its intermediate state. The information about the two fermions samples is listed in Tab.6.

Table 6: The cross section for the 2 fermions background

Process	Cross section [fb]	Error [fb]	ILC result [fb]
uu	9995.35	23.2	10110.43
dd	9808.71	25.7	10010.07
cc	9974.20	28.9	10102.75
ss	9805.39	74.7	9924.40
bb	9803.04	25.5	9957.70
qq	49561.30	120.0	50105.35
e2e2	4967.58	35.0	4991.91
e3e3	4374.94	10.5	4432.18
bhabha	24992.21	41.50	24937.95

It should be noted that the two fermions samples are generated under the averaged polarization.

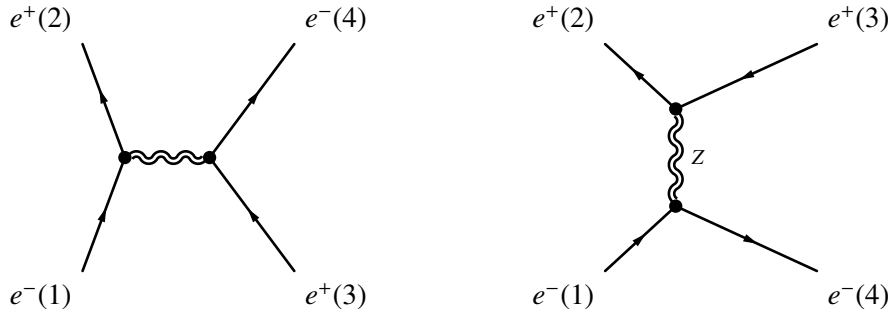


Figure 2: The diagrams of the two fermions processes

4 The four fermions background

The classification of the four fermions background is really a puzzling game, here we will try our best to explain the motivation of such a classification.

First of all, according to the conservation law, the four fermions in the final states can be combined into two bosons, which are Z or W, and the processes can be named as “ZZ” and “WW” respectively. Additionally, when the process of which the final states contains a pair of electron and the accompanying neutrino, it will be excluded from the ZZ and WW groups and named as “single Z” or “single W”, which indicates the origin of the left two fermions. Actually some final particles could come from either “ZZ” or “WW”, for instance, the $\nu_\mu \bar{\nu}_\mu \mu \bar{\mu}$, they will be named as “ZZ or WW mix”. The analogous combination can also happen between the single Z and single W, which will be called as “single Z or single W”.

According to the the classification above, there are six main groups in our hand. Further the samples will be grouped according to content of the final states. The states all consisting of quarks will be

grouped together as “hadronic”, the “semi-leptonic” will stand for the states consisting of two quarks and two leptons, and “leptonic” will be assigned to ones consisting of leptons.

The main topological structure of almost all the four fermions backgrounds Feynman diagrams is what like in Fig.3. The left one is the s -channel, describing the strahlung process, the right one is the t -channel. The internal lines can be any gauge bosons obeying conservation law, not only Z boson. For example, if the fermion involved in Fig.3 are quarks, the line split into two quarks is most likely the gluon due to the large value of strong coupling. The main topological structure of Feynman diagrams for various processes will be given out in Appendix.

It should be noted that there is no direct relationship between the naming schedule and Feynman diagrams. For example, in the “ZZ” processes, actually the final particles do not come from two Z bosons as the name indicates, but from the strahlung process as the Fig.3 shows.

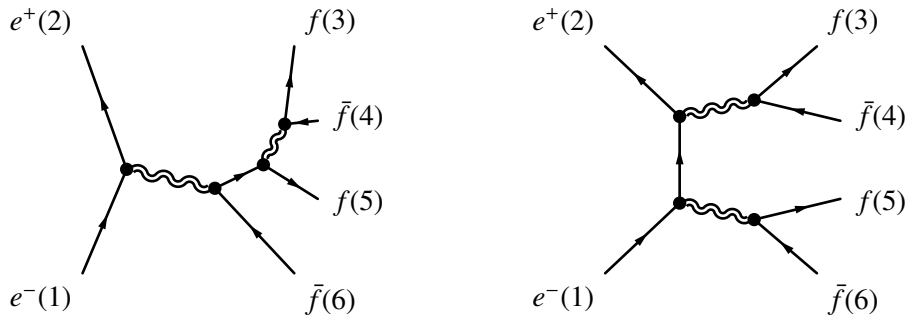


Figure 3: The diagrams for the four fermions processes

4.1 ZZ process

Since the Z boson can be considered as a particle composed of fermion and anti-fermion with the same flavor, the particles in the final states of “ZZ” samples are such combinations of particles with same flavor in principle. However, some of them can also come from the “WW” intermediate process, saying $u\bar{u}d\bar{d}$ e.g., some restrictions will be set to exclude the latter cases and put them into the “ZZ or WW” samples.

First quarks will be put into two groups intuitively, one is named as “up-type”, including the u, c quarks and the their anti partners; the other is the so-called “down-type”, including d, s, b quarks and the antis. To exclude the “WW” intermediate process, the restriction should be set that no flavor changing occurs, so there are four processes left for the hadronic part of the “ZZ” samples, the information of them is shown in Tab.7.

Table 7: The cross section for the hadronic part of ZZ process

Process	Cross section [fb]	Error [fb]	ILC result [fb]
zz_h0utut	83.09	0.112	83.05
zz_h0dtdt	226.20	0.384	226.42
zz_h0uu_notd	95.65	0.119	96.00
zz_h0cc_nots	96.04	0.123	96.28

Here the nameing schedule should be explained, and the same schedule will be used for the other processes of the four fermions background. The “zz” denotes the main combination of the final states as what explained above, there are also “ww”, “sze”, “szn”, “sw”, “zzorww” and “szeorsw” used to

denote the other combinations. It is noted that there are two subgroups for the “single Z”– the “sze” and “szn”, which will be explained more carefully in the Sec.4.3. The character “h” here stands for the word “hadronic”, meanwhile there are “sl” denoting “semi-leptonic” and “l” for “leptonic”. The character “0” is meaningless but just a technical trick used to split the whole string, and the substring at the end is used to indicate the final states and the details can be found in the Sec.5.

The analogous situation should be taken into consideration for the semi-leptonic part of the “ZZ” process. Since the e^+e^- final states has been separated into the single Z samples, the leptons taken into account can be only μ and τ . These two types of leptons with the up and down type quarks, make up four elements of the “ZZ” process. Besides that, the neutrinos of the corresponding leptons make up the left two processes. The information of the semi-leptonic part is given out in Tab.8.

Table 8: The cross section for the semi-leptonic part of ZZ process

Process	Cross section [fb]	Error [fb]	ILC result [fb]
zz_sl0nu_up	81.72	0.091	81.86
zz_sl0tau_down	64.30	0.065	64.64
zz_sl0mu_down	127.96	0.224	128.84
zz_sl0nu_down	134.86	0.166	135.48
zz_sl0tau_up	39.78	0.042	40.02
zz_sl0mu_up	82.38	0.140	83.10

The pure leptonic part of the “ZZ” samples is simpler than the other two parts, the final states are just what in their names. It should be noted that the final states of last two processes, named as “zz_l0mumu” and “zz_l0tautau”, is composed of a pair of lepton and a pair of neutrinos. Take the “zz_l0mumu” as an example, the leptons in the final states are $\mu\bar{\mu}$, and the neutrinos are $\nu_\tau\bar{\nu}_\tau$, since the combination of $\mu\bar{\mu}\nu_\mu\bar{\nu}_\mu$ should be put in the “ZZ or WW” samples. The information of the leptonic part is in Tab.9.

Table 9: The cross section for the leptonic part of ZZ process

Process	Cross section [fb]	Error [fb]	ILC result [fb]
zz_l04tau	4.38	0.005	4.41
zz_l04mu	14.57	0.075	14.63
zz_l0taumu	17.54	0.039	17.73
zz_l0mumu	18.17	0.029	18.34
zz_l0tautau	9.20	0.009	9.27

4.2 WW process

For the “WW” process, there is an additional triple gauge coupling structure Feynman diagram as in Fig.4. The outgoing W bosons tend to leptons and their accompanying neutrinos or the flavor changing quarks for various processes. Unlike the Z boson, W boson can be thought to be composed of two quarks with different flavors, therefore the “WW” samples contain the flavor changing processes. The flavor changing can only occur between the down-type and up-type quarks, after the ones in “ZZ or WW” samples removed, the left hadronic part is given out in Tab.10.

The semi-leptonic final states consist of two quarks and two leptons. The quarks consist of one up-type and one down-type as the hadronic part and the leptons consist of μ or τ and the corresponding

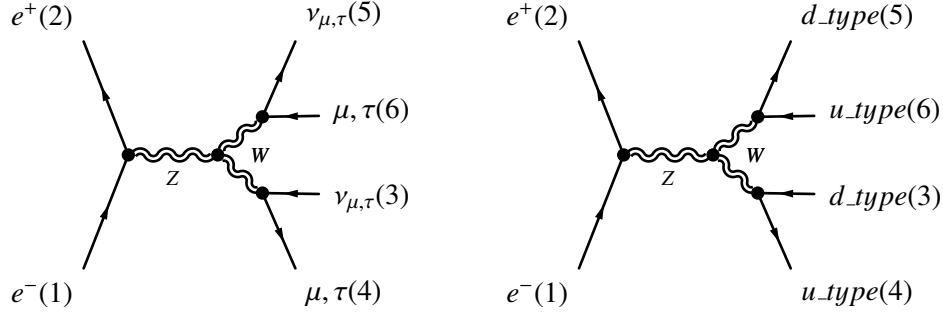


Figure 4: The TGC diagram of the “ww” processes

Table 10: The cross section for the hadronic part of WW process

Process	Cross section [fb]	Error [fb]	ILC result [fb]
ww_h0cuxx	3395.48	4.310	3413.36
ww_h0uubd	0.05	6.37E-5	0.05
ww_h0uusd	165.94	0.209	167.21
ww_h0ccbs	5.74	0.007	5.75
ww_h0ccds	165.57	0.209	166.30

neutrinos since the electron part has been separated into the “single W” samples. Due to the same reason, the final states of the pure leptonic part consist of μ , τ and their neutrinos. The information of these two samples is given out together in Tab.11.

Table 11: The cross section for the semi- and leptonic part of WW process

Process	Cross section [fb]	Error [fb]	ILC result [fb]
ww_sl0tauq	2358.69	2.580	2368.64
ww_sl0muq	2351.98	2.580	2369.79
ww_l0ll	392.96	0.430	394.73

4.3 Single Z process

Since the incoming beams are electron and positron, which makes e^+e^- and $\nu_e\bar{\nu}_e$ doomed to be out of ordinary. Therefore, there are two samples, “single Z” and “single W”, used to simulate the processes in which two final particles have been set to be the electron-relative pair, which means the combination of e^+e^- , or $\nu_e\bar{\nu}_e$, or electron and its accompanying neutrino, in advance. The first two groups belong to the “single Z” process, and named as “single Z-e” and “single Z-nu” respectively, and the last one is the “single W” process.

Addition to the general topological structure of four fermions background, the fusion-like diagrams exist for the so-called “single” processes, including both “single Z” and “single W”. The symbol f in Fig.5 stands for quark and leptons corresponding to the processes concerned.

Unlike the ZZ and WW ones, there are only semi-leptonic and pure leptonic part for them. For the “single Z-e” process, the quarks in the semi-leptonic part are only up-type or down-type, and leptons in the leptonic are μ , τ and their accompanying neutrinos. The “single Z-e” is listed in Tab.12.

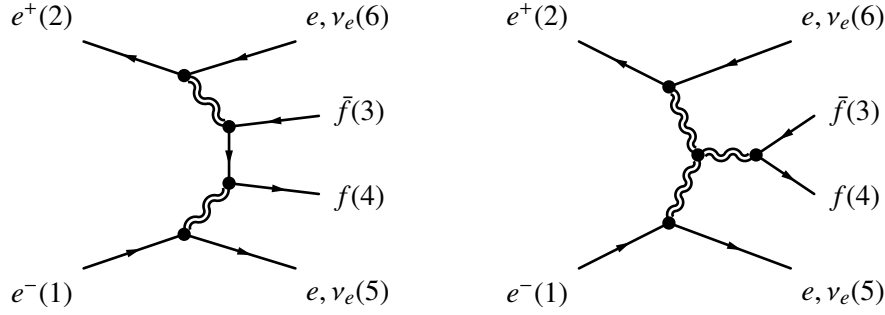


Figure 5: The diagrams for the four fermions processes

Table 12: The cross section for the SZE process

Process	Cross section [fb]	Error [fb]	ILC result [fb]
sze_l0tau	150.14	0.366	150.30
sze_l0mu	852.18	2.400	850.70
sze_l0nunu	29.62	0.036	29.41
sze_sl0uu	195.86	0.441	195.37
sze_sl0dd	128.72	0.191	128.20

Copying the structure of the “single Z-e” without the neutrino particle due to the fact it is invisible, we produce the “single Z-nu” samples. The information is given out in Tab.13

Table 13: The cross section for the SZNU process

Process	Cross section [fb]	Error [fb]	ILC result [fb]
sznu_l0mumu	43.33	0.085	43.41
sznu_l0tautau	14.57	0.018	14.61
sznu_sl0nu_up	56.09	0.069	55.95
sznu_sl0nu_down	91.28	0.118	90.95

4.4 Single W process

The “single W” sample is simpler than “single Z”. If the left two particles are quarks, they will be up-type and down-type one by one definitely; otherwise, they will be μ or τ and the corresponding neutrinos for the pure leptonic part.

Table 14: The cross section for the single W process

Process	Cross section [fb]	Error [fb]	ILC result [fb]
sw_l0mu	429.20	0.506	430.64
sw_l0tau	429.42	0.511	430.27
sw_sl0qq	2579.31	3.020	2581.03

4.5 Z or W mixing process

There are some combinations of particles could come from both Z and W boson, the corresponding processes form the mixing samples. The hadronic part consists of two processes, the final states of process “h0udud” consist of two u quarks and two d quarks, meanwhile the the final states of process “h0cscs” consist of two c quarks and two s quarks. The final particles of the three leptonic ones are the three generations of leptons and their accompanying neutrinos. The information for the mixing samples is given out in Tab.15.

Table 15: The cross section for the mixed process

Process	Cross section [fb]	Error [fb]	ILC result [fb]
zzorww_h0udud	1570.40	2.120	1579.63
zzorww_h0cscs	1568.94	2.100	1572.41
zzorww_l0mumu	214.81	0.272	216.12
zzorww_l0tautau	205.84	0.246	206.38
szeorsw_l0l	249.34	0.346	249.74

5 Conclusion

At the end of this note, all the information given above is collected in Tab.17-Tab.19. The alias used to shorten the table is listed in Tab.16.

Table 16: The alias for particles

$uq : u, \bar{u}$	$up : u, \bar{u}, c, \bar{c}$	$nu_e : \nu_e, \bar{\nu}_e$
$dq : d, \bar{d}$	$down : d, \bar{d}, s, \bar{s}, b, \bar{b}$	$nu_\mu : \nu_\mu, \bar{\nu}_\mu$
$cq : c, \bar{c}$	$e : e^-, e^+$	$nu_\tau : \nu_\tau, \bar{\nu}_\tau$
$sq : s, \bar{s}$	$mu : \mu^-, \mu^+$	$nu_{\mu,\tau} : \nu_{\mu,\tau}, \bar{\nu}_{\mu,\tau}$
$bq : b, \bar{b}$	$tau : \tau^-, \tau^+$	$nu : \nu_{e,\mu,\tau}, \bar{\nu}_{e,\mu,\tau}$
$f : e^-, \mu, \tau, \nu_e, \nu_\mu, \nu_\tau, u, d, c, s, b$	$q : u, d, c, s, b$	

Table 17: The information of the Higgs signal samples

Process	Final states	σ [fb]	ILC result [fb]	Events expected	Events generated
ffh_X	$h, f, \bar{f}$	211.01	208.77	1065619	1065111
qqh_X	$h, q, \bar{q}$	143.39	141.99	724097	723755
uuh_X	$h, u, \bar{u}$	24.52	–	123802	123733
ddh_X	$h, d, \bar{d}$	31.45	–	158830	158742
cch_X	$h, c, \bar{c}$	24.51	–	123766	123711
ssh_X	$h, s, \bar{s}$	31.46	–	158891	158803
bbh_X	$h, b, \bar{b}$	31.18	–	157479	157412
e1e1h_X	h, e^-, e^+	7.60	7.19	38357	99938
e2e2h_X	h, μ^-, μ^+	7.10	7.03	35849	99952
e3e3h_X	h, τ^-, τ^+	7.08	7.02	35770	99951
nnh_X	$h, \nu_{e,\mu,\tau}, \bar{\nu}_{e,\mu,\tau}$	48.96	48.43	247273	247167
n1n1h_X	$h, \nu_e, \bar{\nu}_e$	20.91	–	105574	105533
n2n2h_X	$h, \nu_\mu, \bar{\nu}_\mu$	14.03	–	70862	99961
n3n3h_X	$h, \nu_\tau, \bar{\nu}_\tau$	14.01	–	70773	99944

Table 18: The information of the two fermions background samples

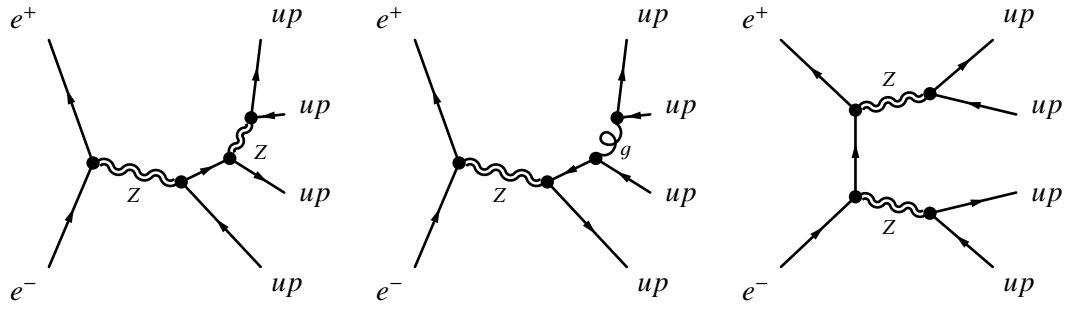
Process	Final states	σ [fb]	ILC result [fb]	Events expected	Events generated
uu	$u, \bar{u}$	9995.35	10110.43	50476527	50476526
dd	$d, \bar{d}$	9808.71	10010.07	49533965	49533961
cc	$c, \bar{c}$	9974.20	10102.75	50369725	50369718
ss	$s, \bar{s}$	9805.39	9924.40	49517234	49517231
bb	$b, \bar{b}$	9803.04	9957.70	49505372	49504516
qq	$q, \bar{q}$	49561.30	50105.35	250284565	250283714
e2e2	μ^-, μ^+	4967.58	4991.91	25086253	25086255
e3e3	τ^-, τ^+	4374.94	4432.18	22093447	22093445
bhabha	e^-, e^+, γ	24992.21	24937.95	126210660	126210654

Table 19: The information of the four fermions background samples

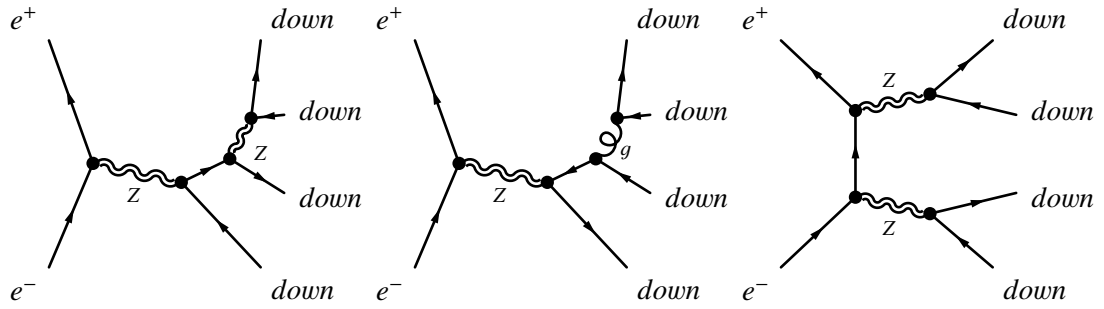
Process	Final states	σ [fb]	ILC result [fb]	Events expected	Events generated
zz_h0utut	up, up, up, up	83.09	83.05	419604	419584
zz_h0dtdt	$down, down, down, down$	226.20	226.42	1142310	1142270
zz_h0uu_notd	$uq, uq, (sq, bq), (sq, bq)$	95.65	96.00	483032	483045
zz_h0cc_nots	$cq, cq, (dq, bq), (dq, bq)$	96.04	96.28	485002	485016
zz_sl0nu_up	$\nu_{\mu,\tau}, \nu_{\mu,\tau}, up, up$	81.72	81.86	412686	412709
zz_sl0nu_down	$\nu_{\mu,\tau}, \nu_{\mu,\tau}, down, down$	134.86	135.48	681043	681041
zz_sl0mu_up	μ, μ, up, up	82.38	83.10	416019	416008
zz_sl0mu_down	$\mu, \mu, down, down$	127.96	128.84	646198	646181
zz_sl0tau_up	τ, τ, up, up	39.78	40.02	200889	200882
zz_sl0tau_down	$\tau, \tau, down, down$	64.30	64.64	324715	324709
zz_l04tau	$\tau^-, \tau^+, \tau^-, \tau^+$	4.38	4.41	22119	100000
zz_l04mu	$\mu^-, \mu^+, \mu^-, \mu^+$	14.57	14.63	73578	100000
zz_l0taumu	$\tau^-, \tau^+, \mu^-, \mu^+$	17.54	17.73	88577	100000
zz_l0mumu	$\nu_\tau, \bar{\nu}_\tau, \mu^-, \mu^+$	18.17	18.34	91758	100000
zz_l0tautau	$\nu_\mu, \bar{\nu}_\mu, \tau^-, \tau^+$	9.20	9.27	46460	100000
ww_h0cuxx	$uq, cq, down, down$	3395.48	3413.36	17147189	17147188
ww_h0uubd	uq, uq, dq, bq	0.05	0.05	252	100000
ww_h0uusd	uq, uq, sq, bq	165.94	167.21	837997	838010
ww_h0ccbbs	cq, cq, sq, bq	5.74	5.75	28987	100000
ww_h0ccds	cq, cq, sq, dq	165.57	166.30	836128	836128
ww_sl0muq	$\mu, \nu, up, down$	2358.69	2369.79	11911394	11911396
ww_sl0tauq	$\tau, \nu, up, down$	2351.98	2368.64	11877519	11877519
ww_l0ll	$\mu, \tau, \nu_\mu, \nu_\tau$	392.96	394.73	1984448	1984437
zzorww_h0udud	uq, uq, dq, dq	1570.40	1579.63	7930514	7930514
zzorww_h0cscs	cq, cq, sq, sq	1568.94	1572.41	7923141	7923140
zzorww_l0mumu	$\mu, \mu, \nu_\mu, \nu_\mu$	214.81	216.12	1084790	1084777
zzorww_l0tautau	$\tau, \tau, \nu_\tau, \nu_\tau$	205.84	206.38	1039492	1039510
sze_l0tau	e^-, e^+, τ^-, τ^+	150.14	150.30	758207	758206
sze_l0mu	e^-, e^+, μ^-, μ^+	852.18	850.70	4303527	4303528
sze_l0nnu	$e^-, e^+, \nu_{\mu,\tau}, \bar{\nu}_{\mu,\tau}$	29.62	29.41	149581	149583
sze_sl0uu	e, e, up, up	195.86	195.37	989093	989109
sze_sl0dd	$e, e, down, down$	128.72	128.20	650036	649940
sznu_l0mumu	$\nu_e, \bar{\nu}_e, \mu^-, \mu^+$	43.33	43.41	218816	218824
sznu_l0tautau	$\nu_e, \bar{\nu}_e, \tau^-, \tau^+$	14.57	14.61	73578	100000
sznu_sl0nu_up	$\nu_e, \bar{\nu}_e, up, up$	56.09	55.95	283254	283254
sznu_sl0nu_down	$\nu_e, \bar{\nu}_e, down, down$	91.28	90.95	460964	460961
sw_l0mu	$e, \nu_e, \mu, \nu_{\mu,\tau}$	429.20	430.64	2167446	2167447
sw_l0tau	$e, \nu_e, \tau, \nu_{\mu,\tau}$	429.42	430.27	2168556	2168556
sw_sl0qq	$e, \nu_e, up, down$	2579.31	2581.03	13025535	13025535
szeorsw_l0l	$e^-, e^+, \nu_e, \bar{\nu}_e$	249.34	249.74	1259167	1259165

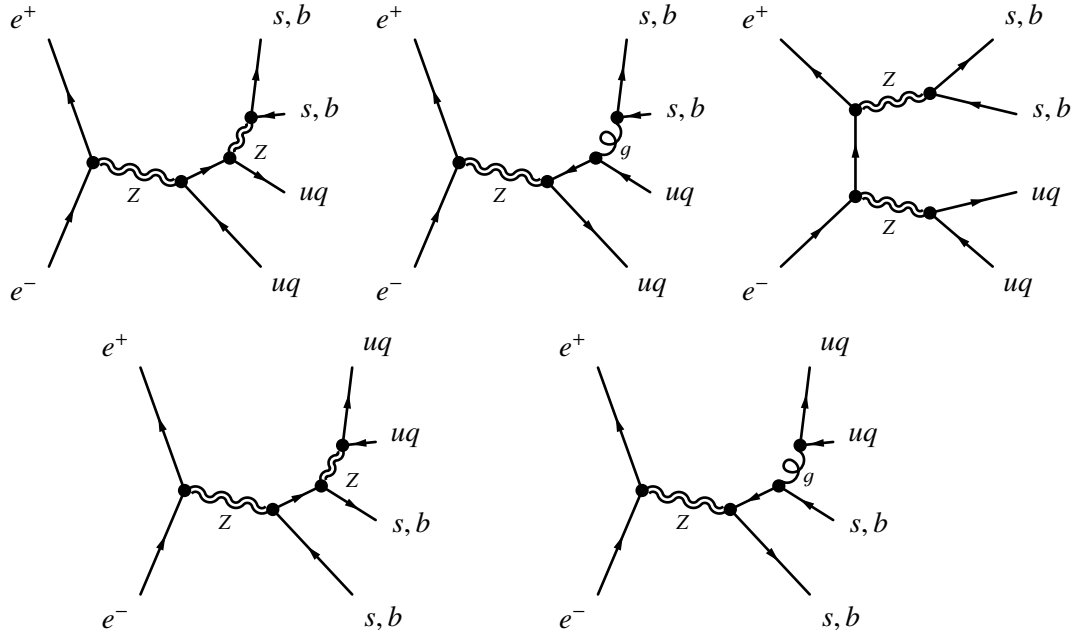
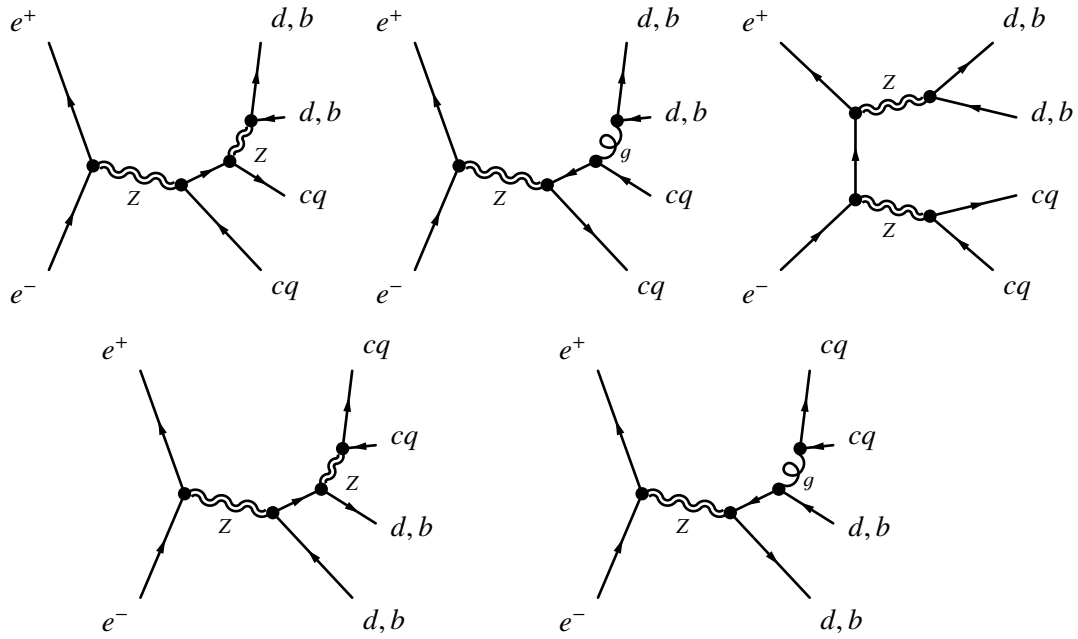
180 6 Appendix: Feynman diagrams for 4 fermions background

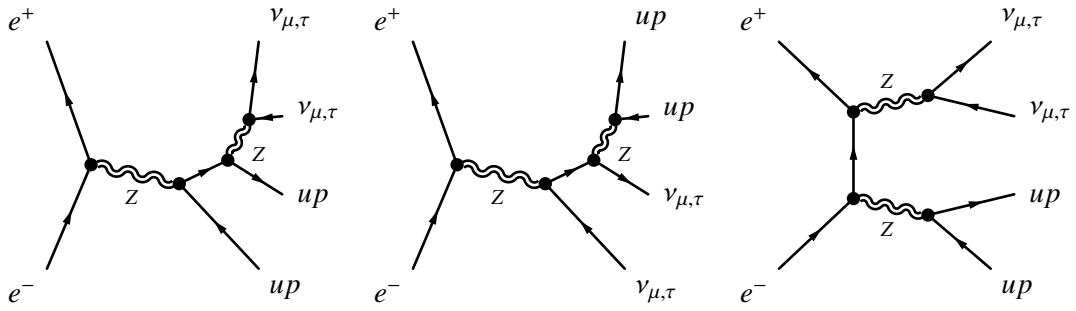
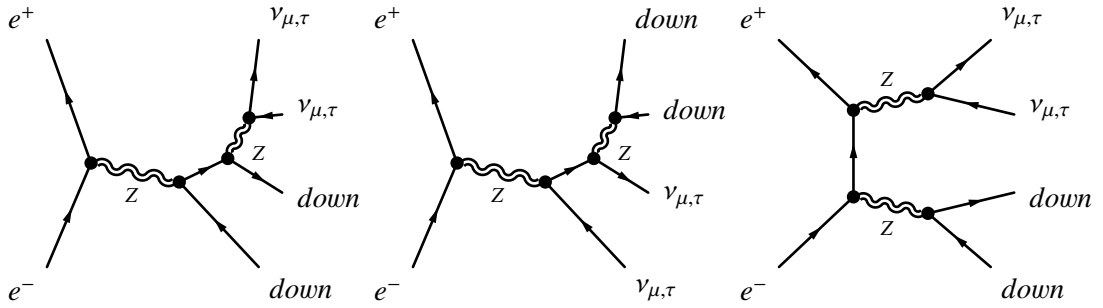
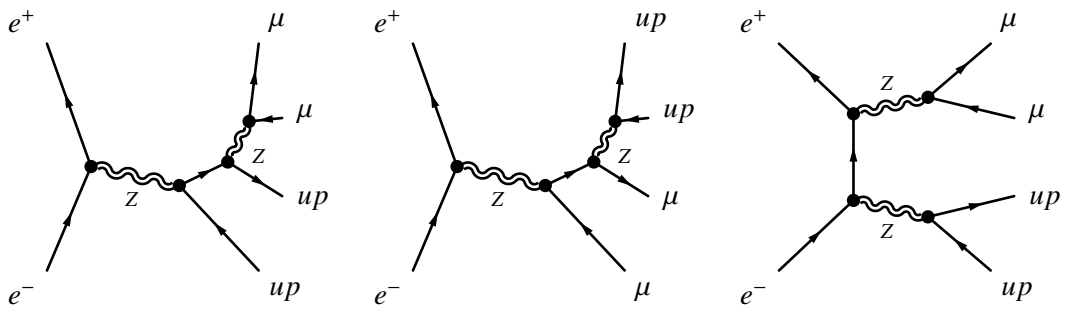
181 6.1 zz_h0utut

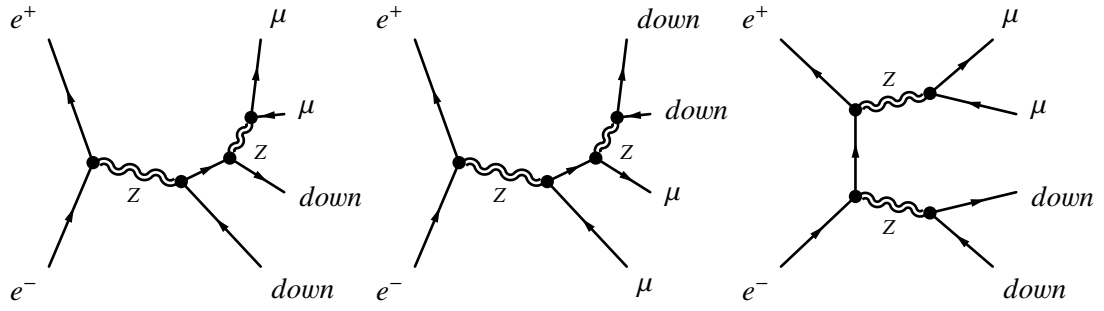
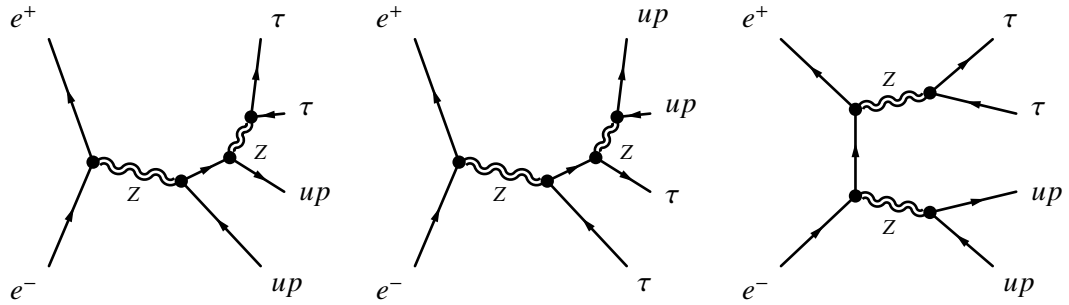
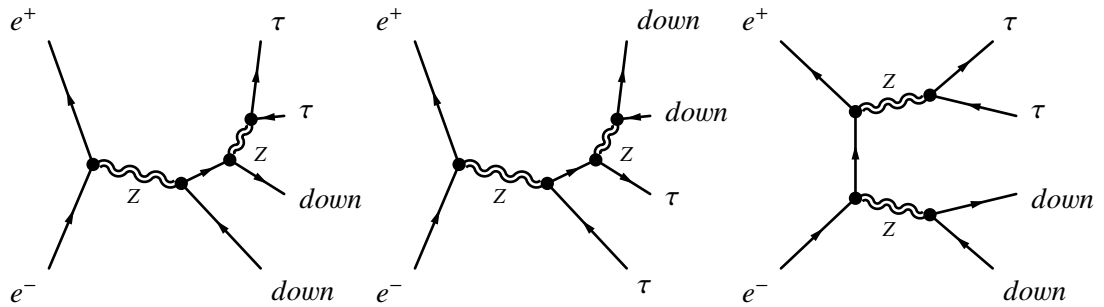


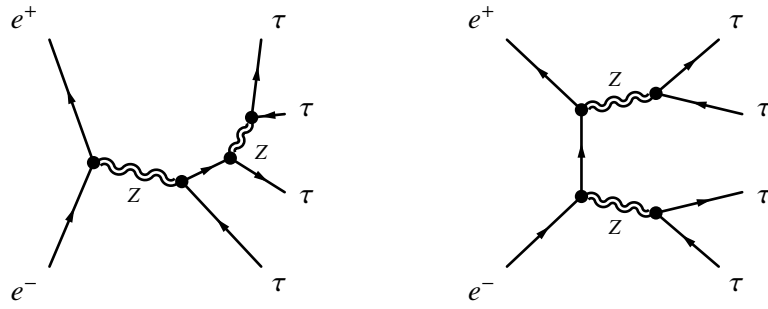
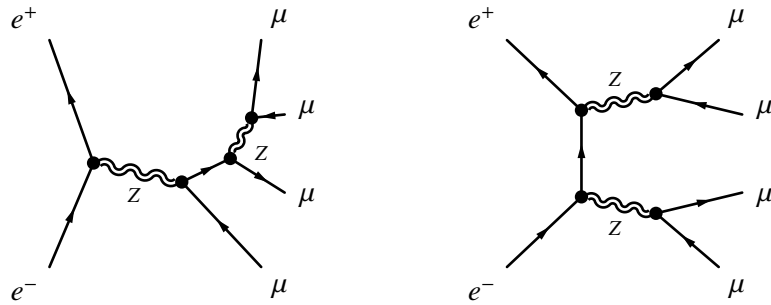
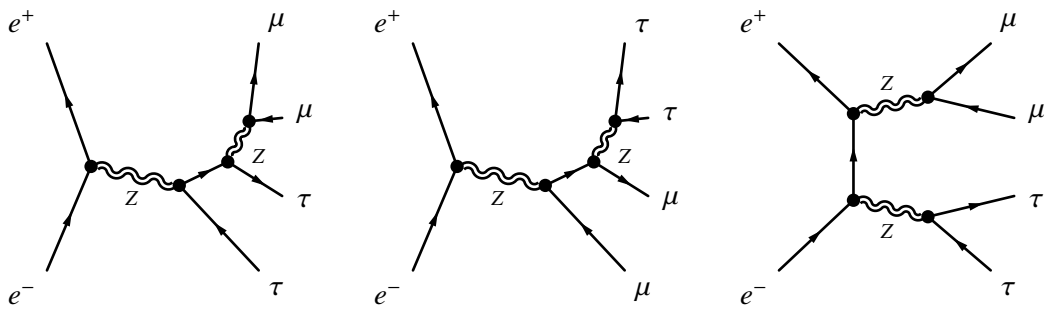
182 6.2 zz_h0dtdt

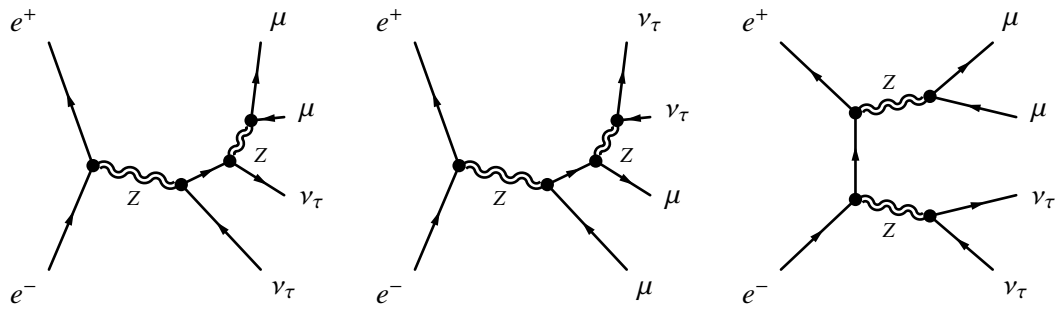
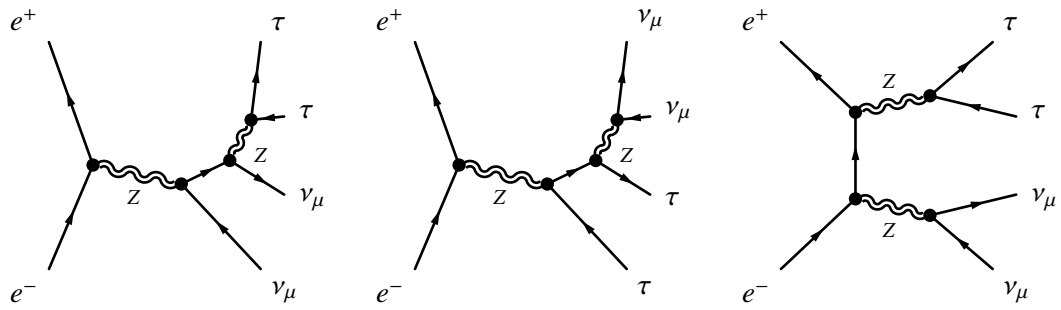


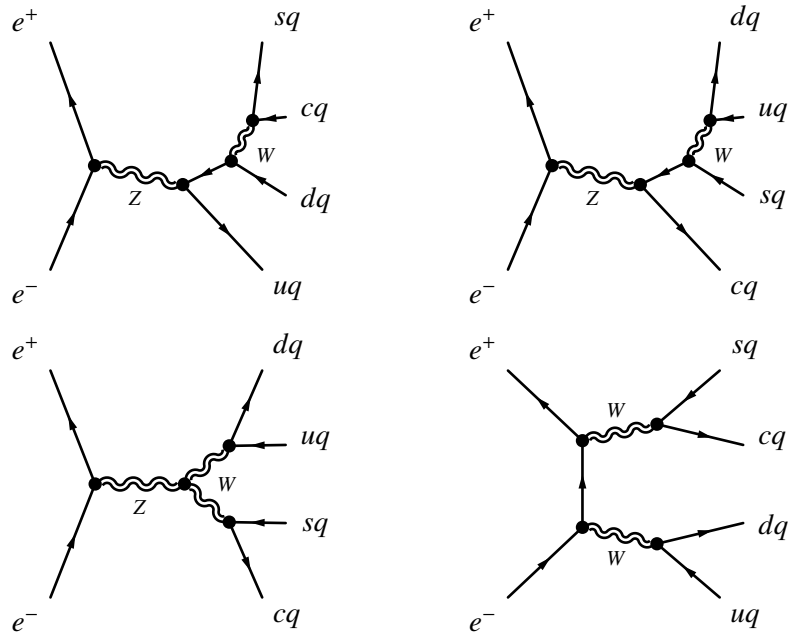
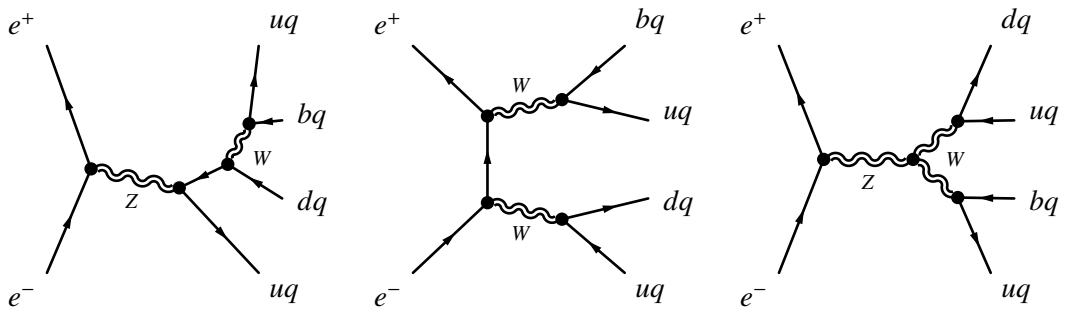
183 **6.3 zz_h0uu_notd**184 **6.4 zz_h0cc_notd**

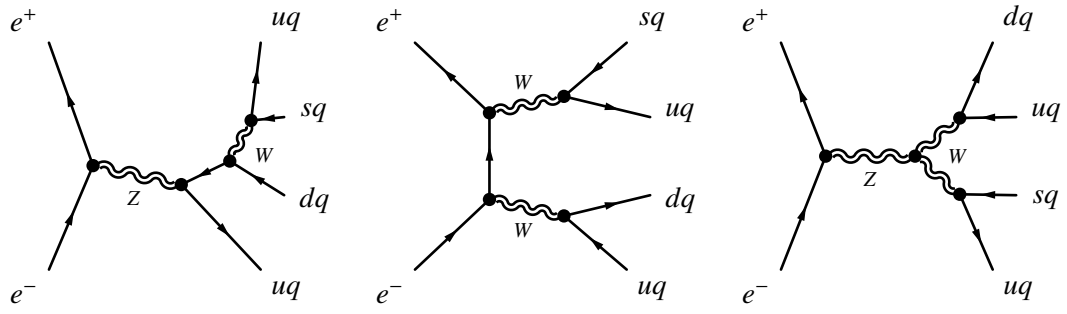
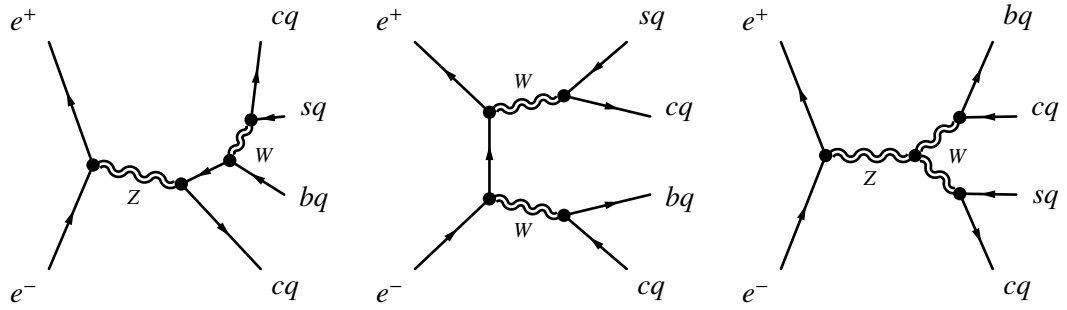
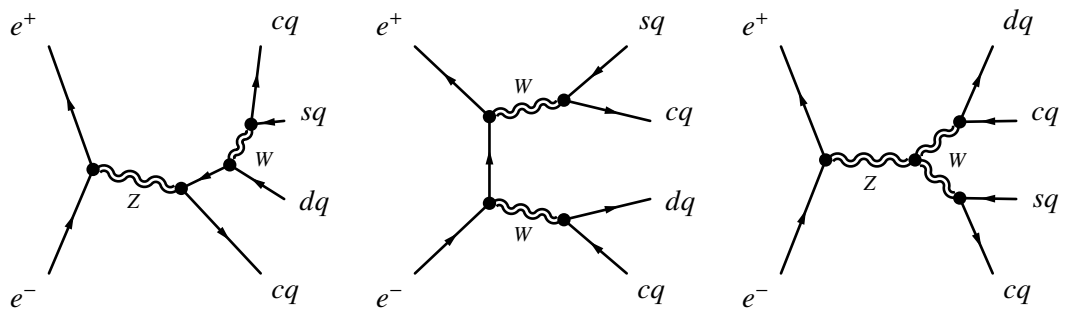
185 **6.5** `zz_sl0nu_up`186 **6.6** `zz_sl0nu_down`187 **6.7** `zz_sl0mu_up`

188 **6.8** `zz_sl0mu_down`189 **6.9** `zz_sl0tau_up`190 **6.10** `zz_sl0tau_down`

191 **6.11 zz_l04tau**192 **6.12 zz_l04mu**193 **6.13 zz_l0taumu**

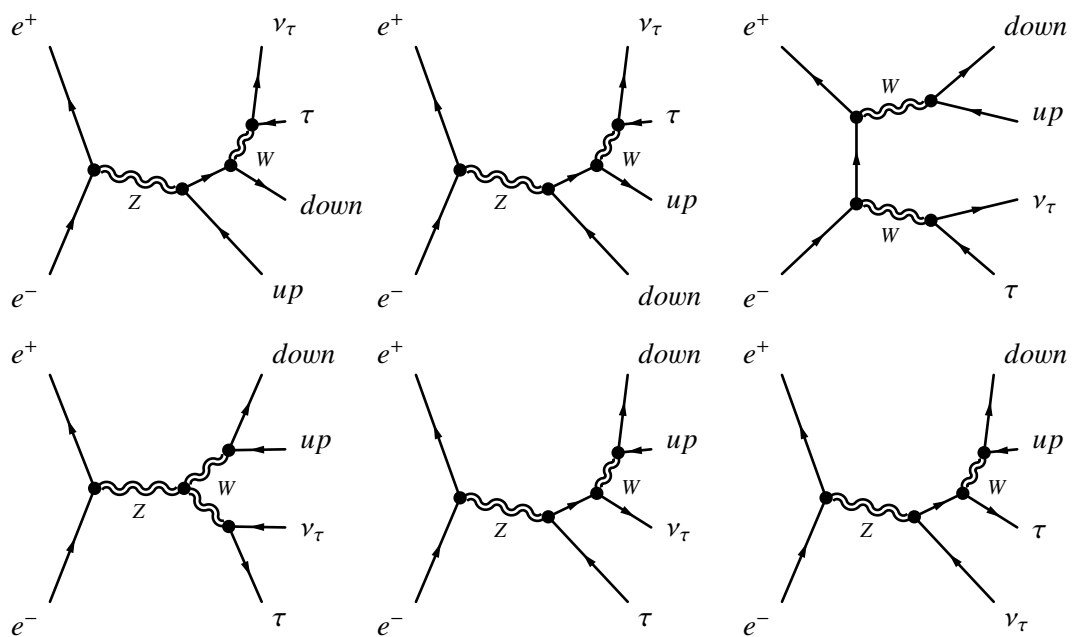
194 **6.14** **zz_l0mumu**195 **6.15** **zz_l0tautau**

196 **6.16** **ww_h0cuxx**197 **6.17** **ww_h0uubd**

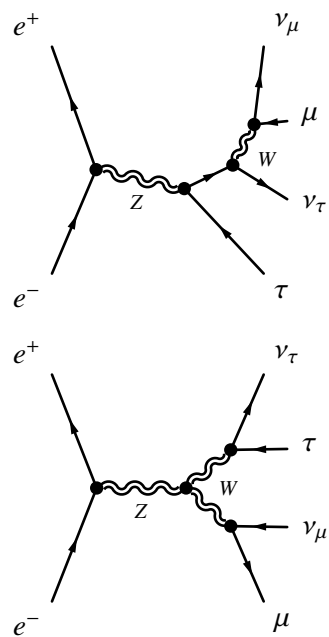
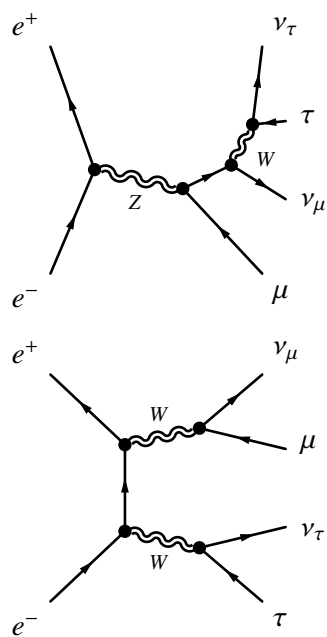
198 **6.18** **ww_h0uusd**199 **6.19** **ww_h0ccbs**200 **6.20** **ww_h0ccds**

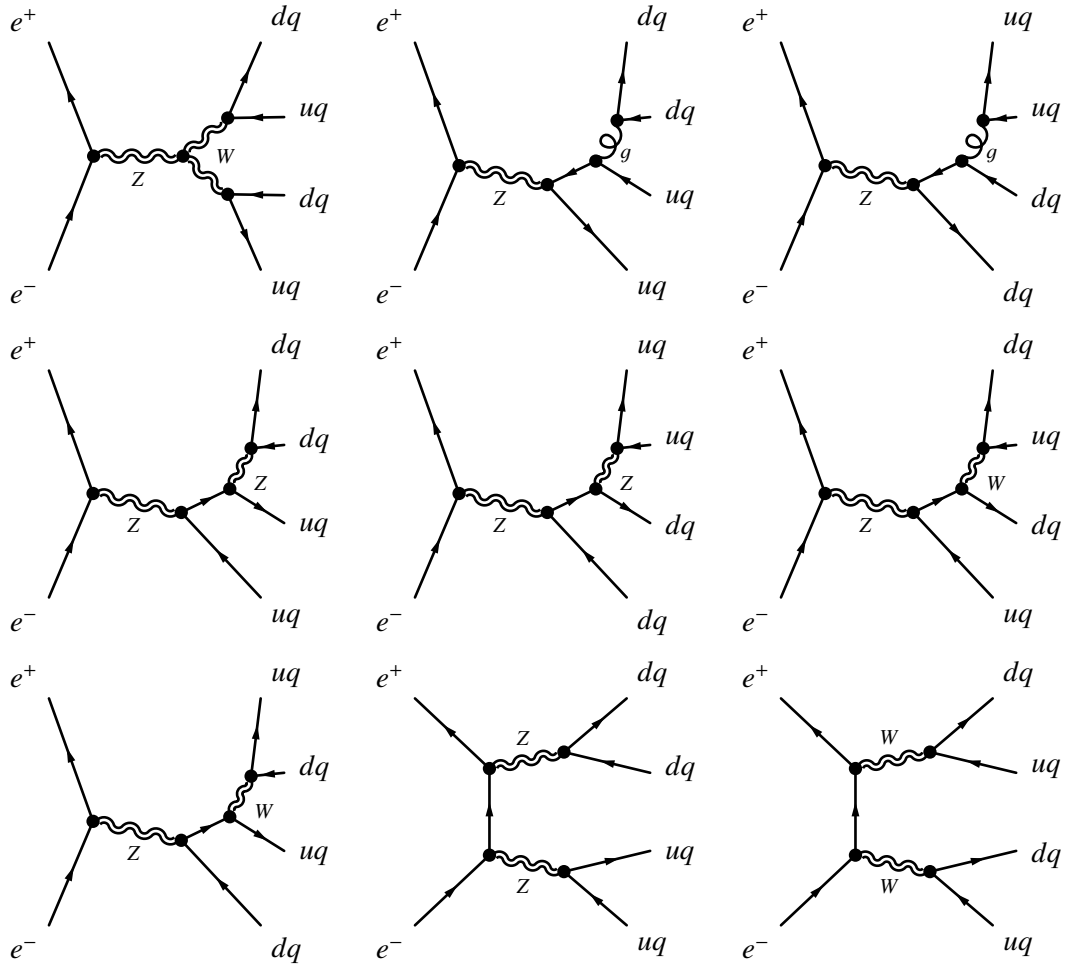
201

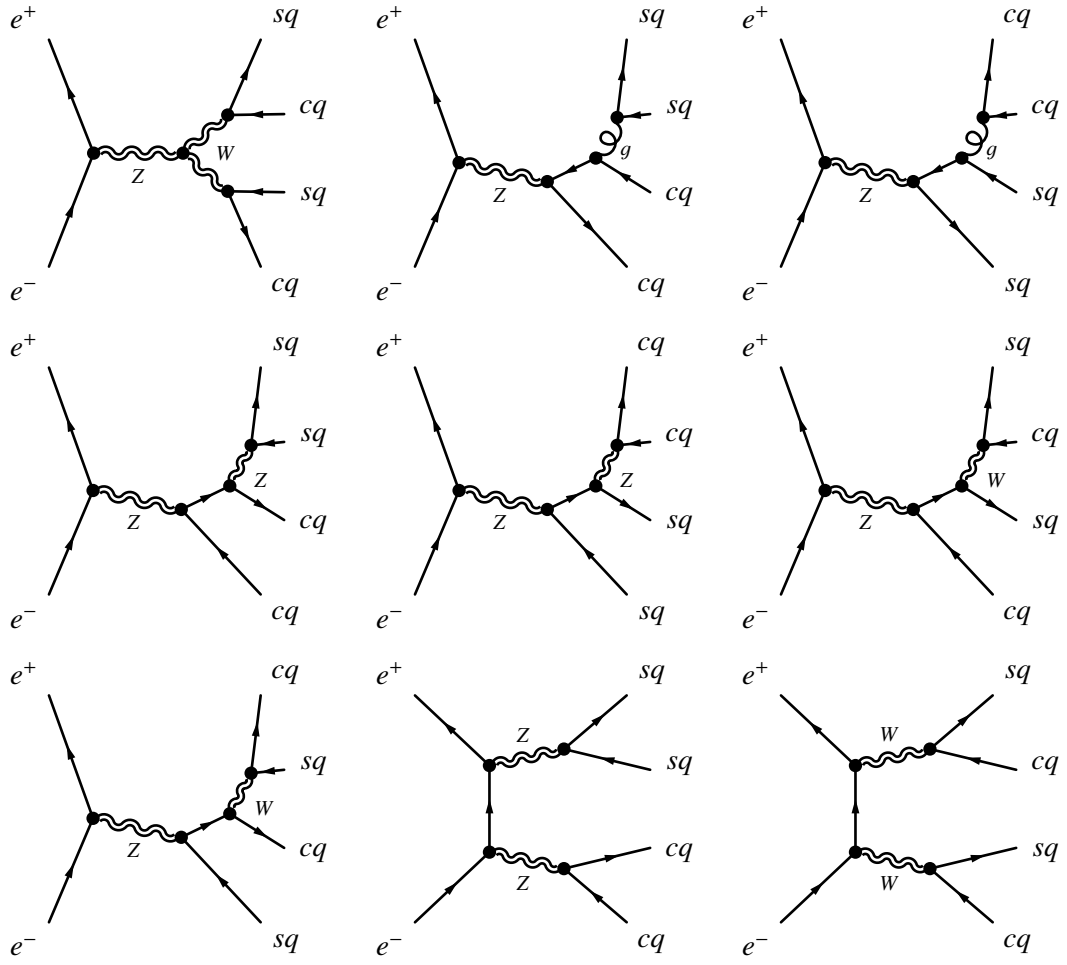
202



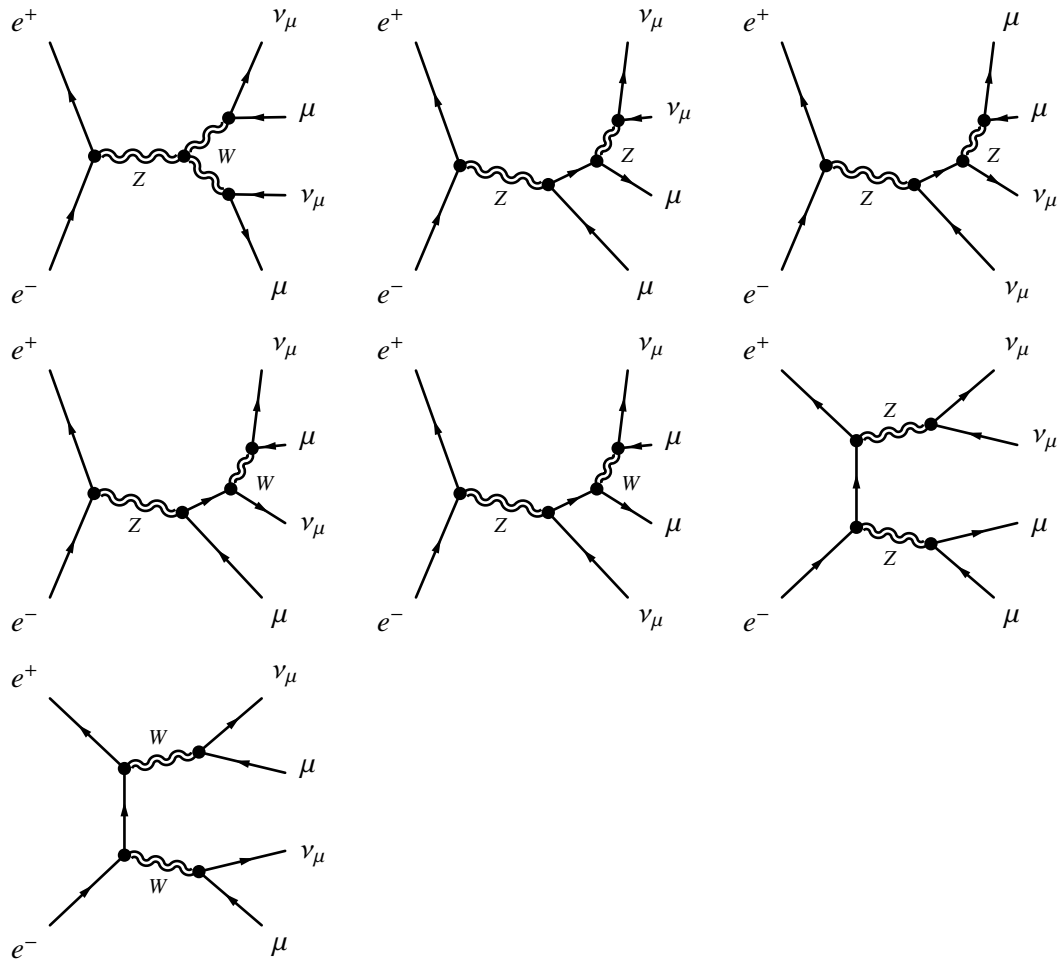
203 **6.23** ww_1011

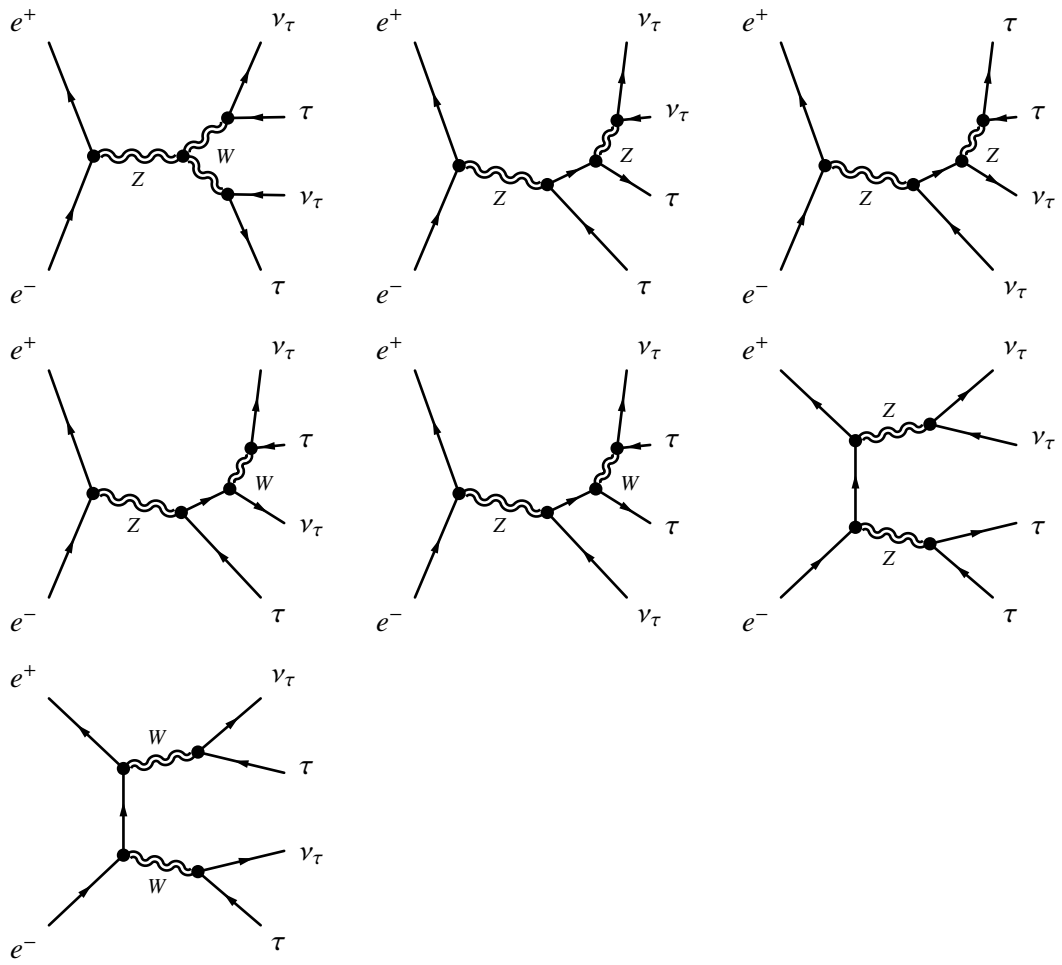


204 **6.24** zzorww_h0udud

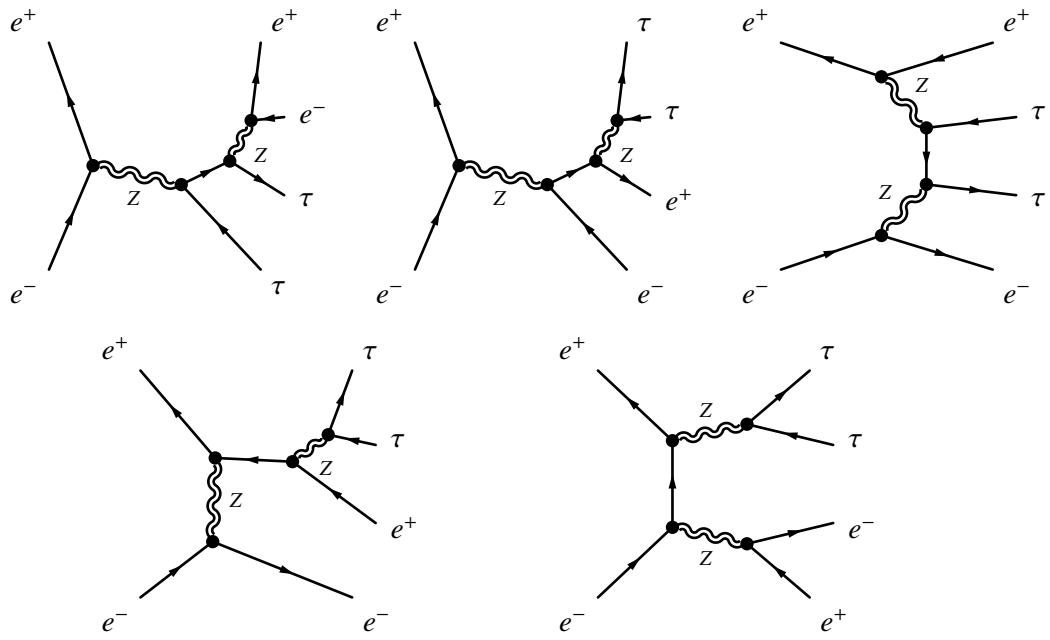
205 **6.25** `zzorww_h0cscs`

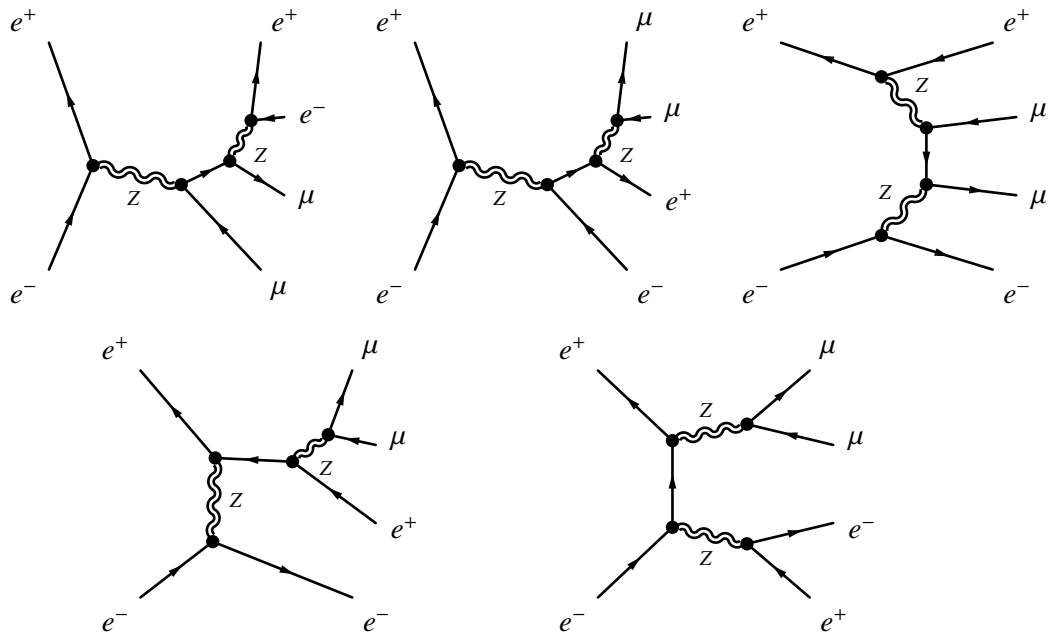
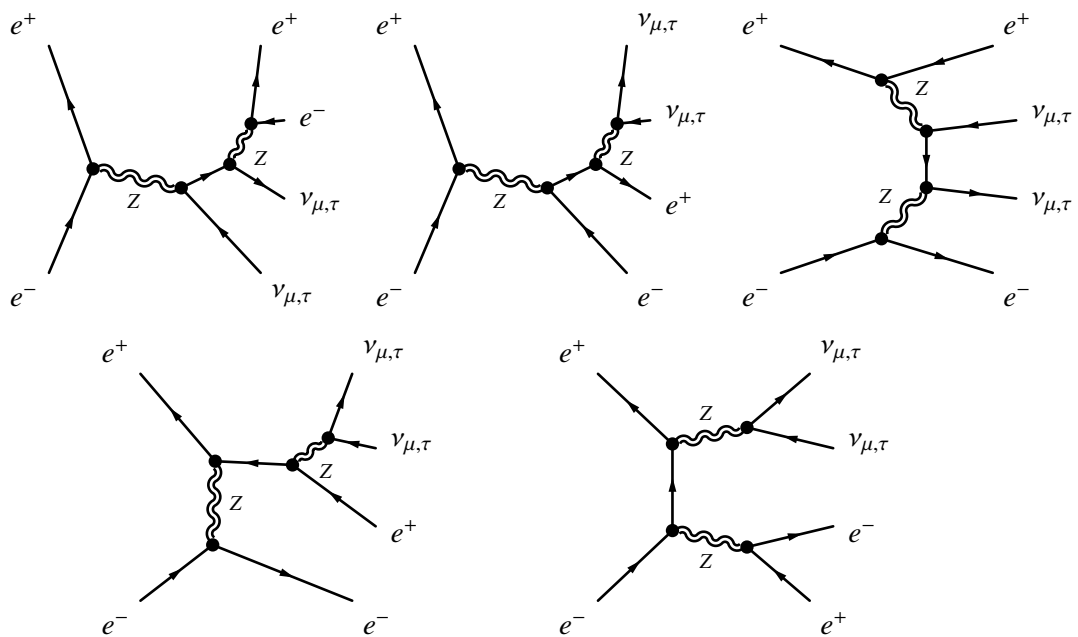
206 6.26 zzorww_l0mumu

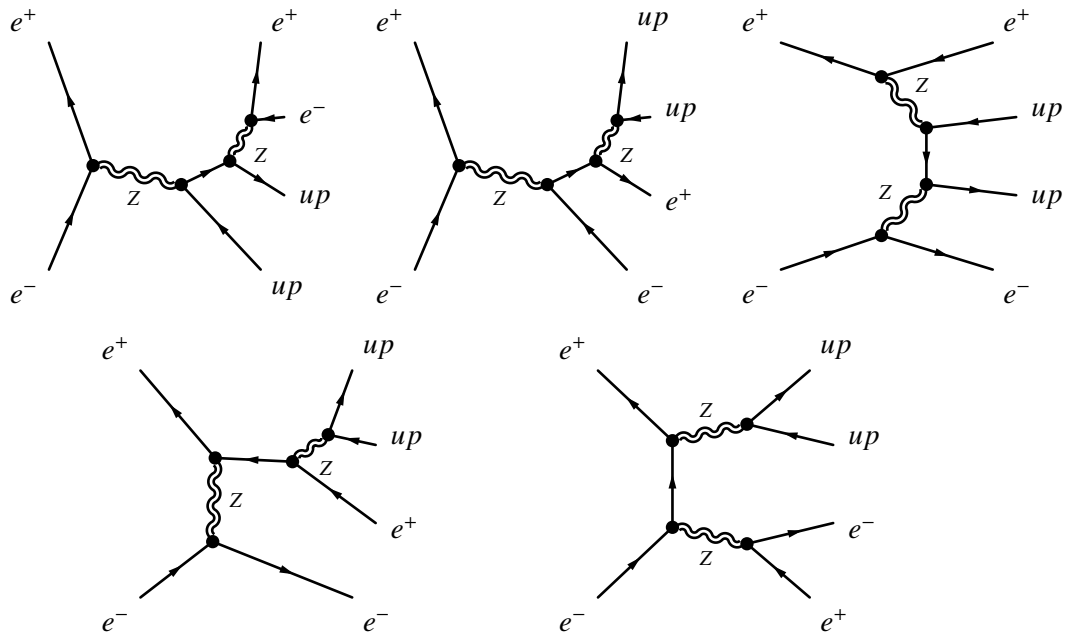
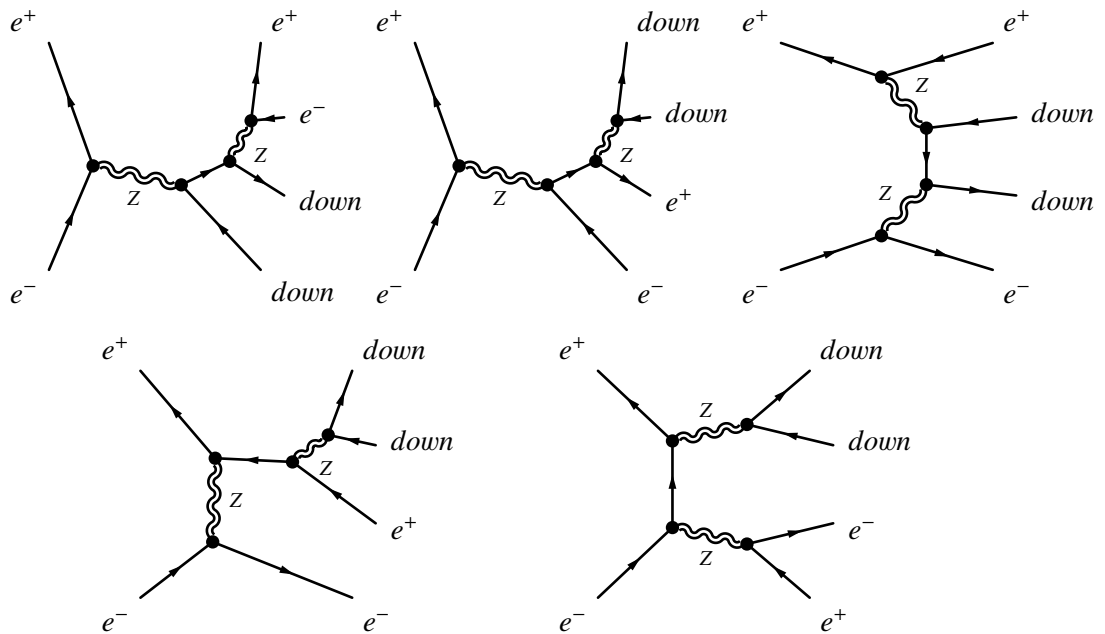


207 **6.27** zzorww_l0tautau

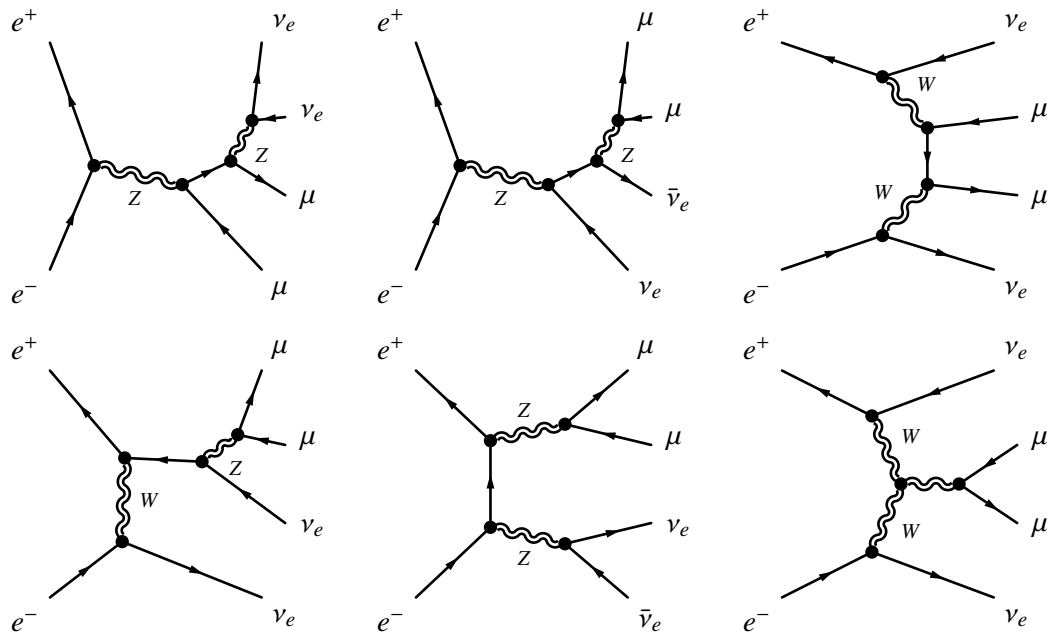
208 6.28 sze_l0tau



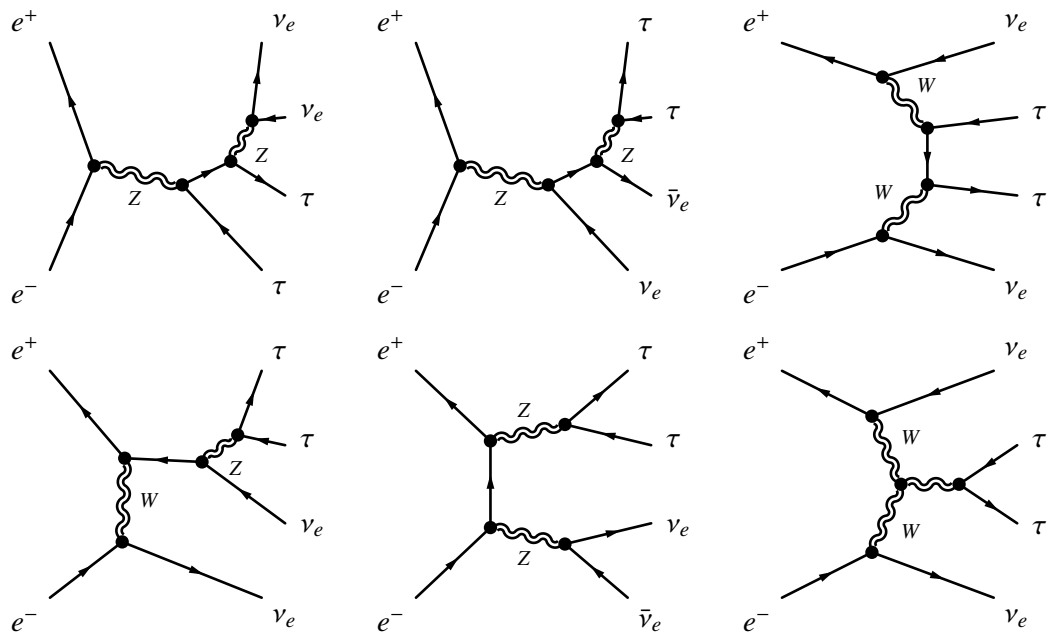
209 **6.29** `sze_l0mu`210 **6.30** `sze_l0nunu`

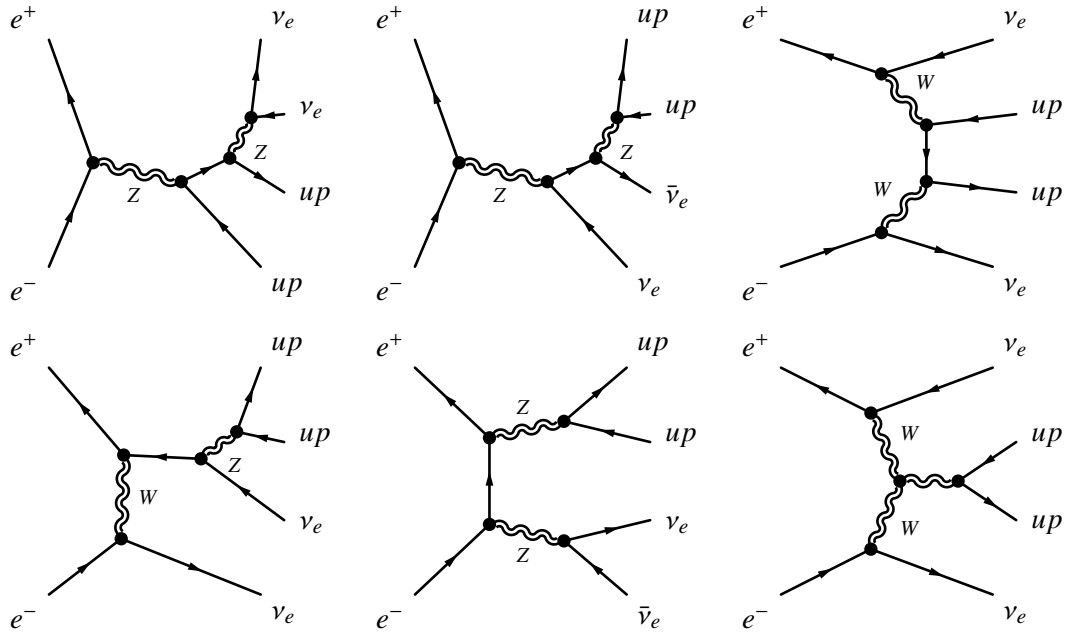
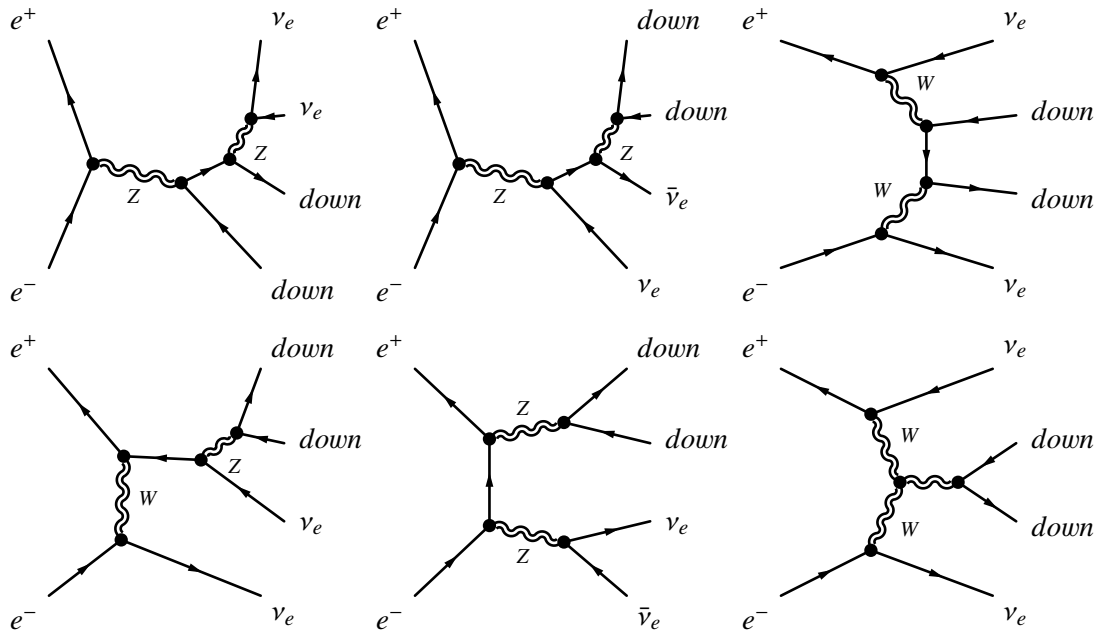
211 **6.31** `size_sl0uu`212 **6.32** `size_sl0dd`

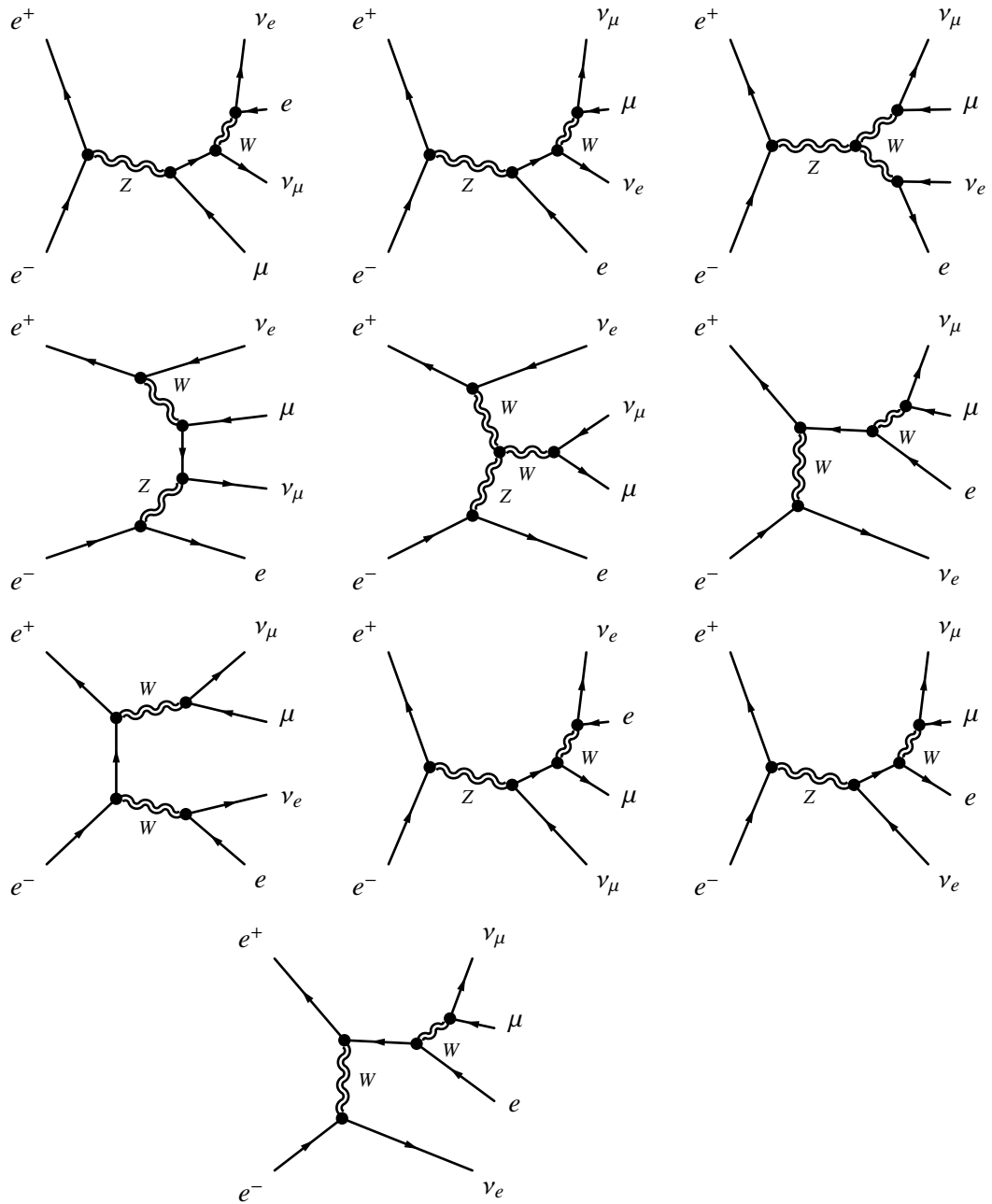
213 6.33 sznu_l0mumu



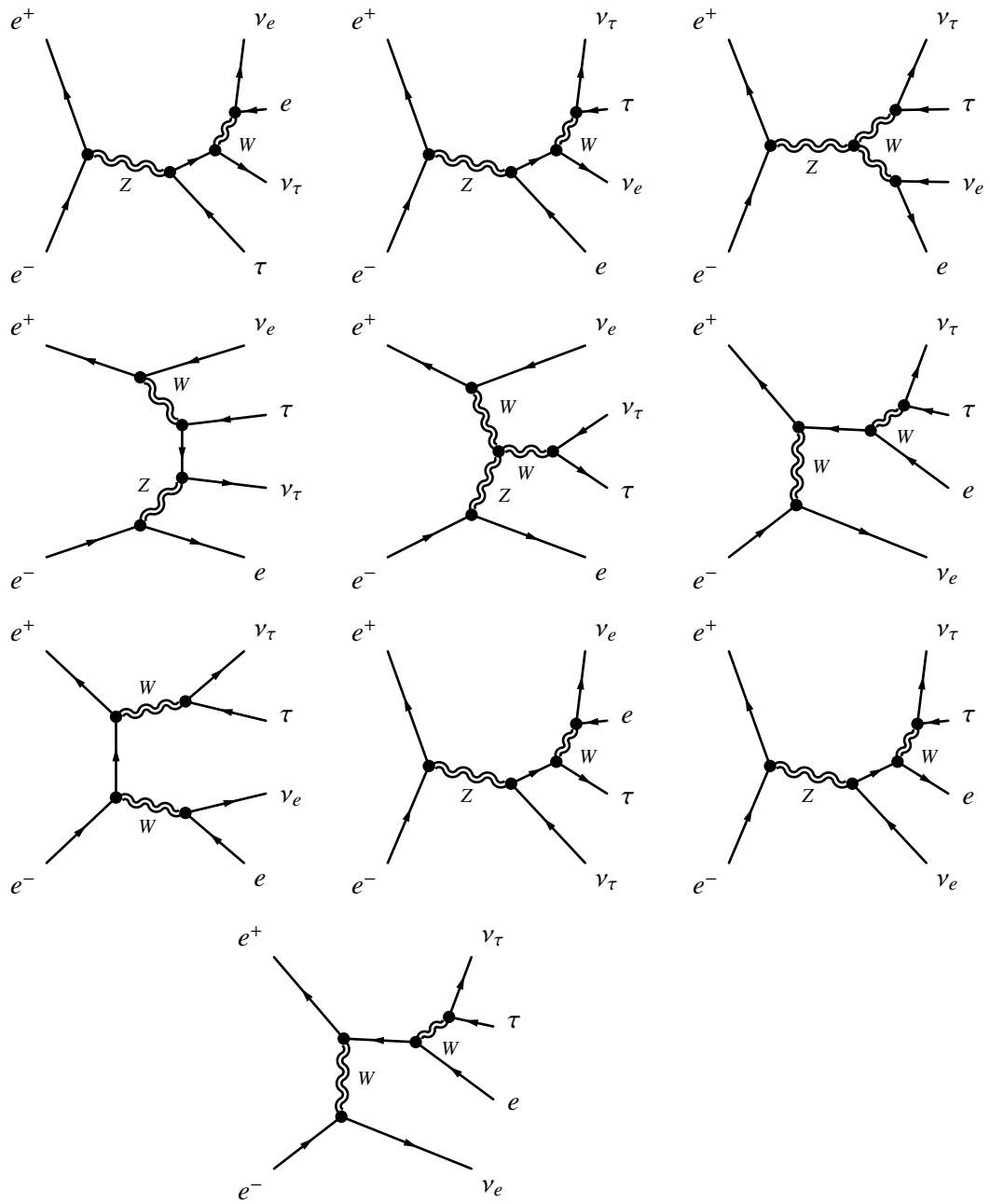
214 6.34 sznu_l0tautau

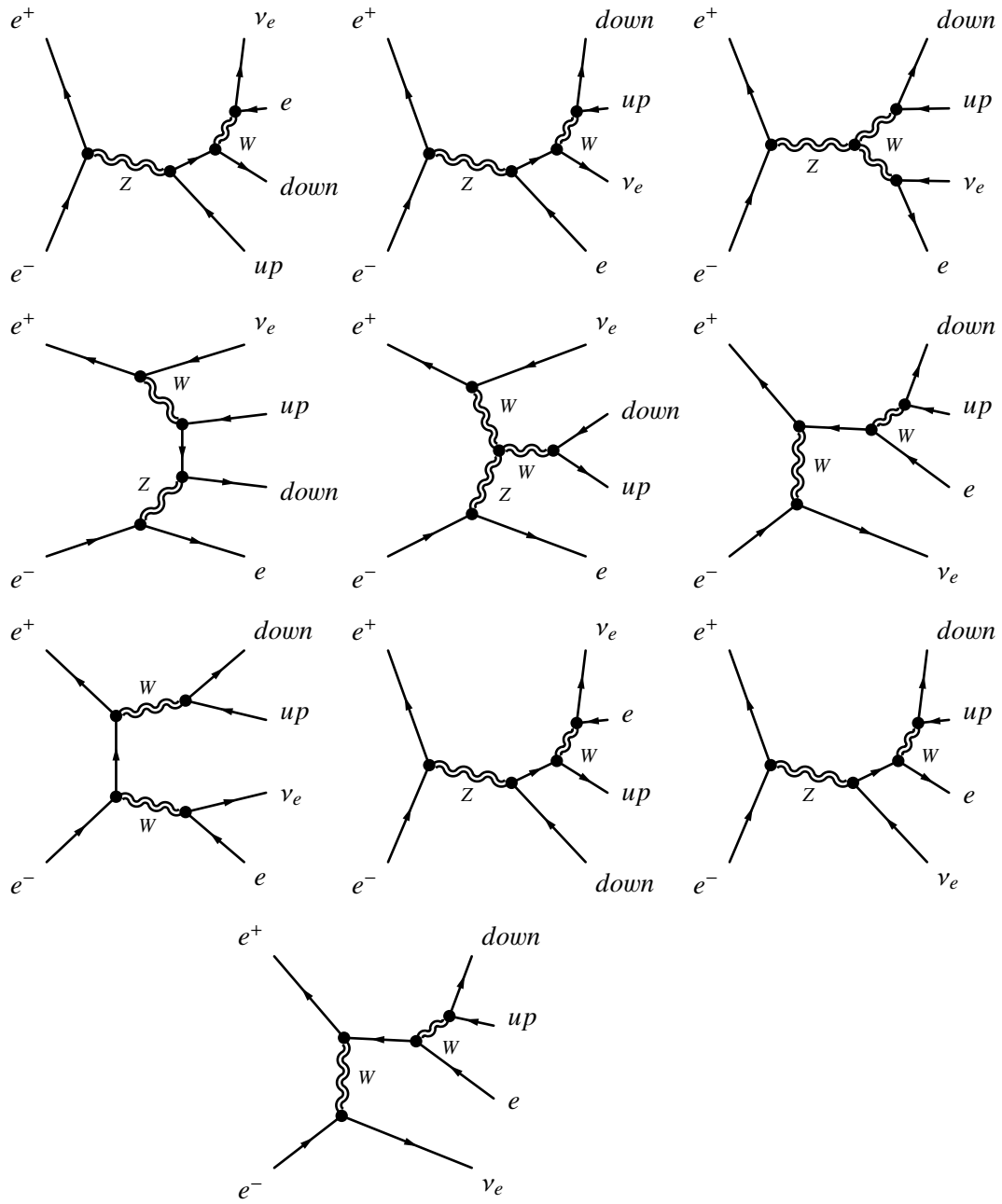


215 **6.35** sznu_sl0nu_up216 **6.36** sznu_sl0nu_down

217 **6.37** sw_l0mu

218 **6.38** sw_l0tau



219 **6.39** sw_sl0qq

220 6.40 szeorsw_l01

